

DWDM Practice Questions

1. Generate association rules from following data set using Apriori Algorithm (Assume support= 40% and confidence= 65%)

ID	Items
1	A B D
2	B D E
3	A B D E
4	B C E
5	B D E

2. Compute frequent itemsets from following data set using FP-Tree and Apriori algorithm. (Assume min_support=3)

TID	Items
T1	A, C, T, W
T2	C, D, W
T3	C, A, T, W
T4	W, A, C, D
T5	A, C, T, D, W
T6	C, D, T

3. Compute frequent itemsets from following data set using FP-Tree. (Assume min_support=3)

TID	Items
T1	M, O, N, K, E, Y
T2	D, O, N, K, E, Y
T3	M, A, K, E
T4	M, U, C, K Y
T5	C, O, O, K, I, E

4. Generate association rules from following data set using FP-Tree (Assume support= 30% and confidence= 65%)

Transaction Data Set

TID	Items
1	{a,b}
2	{b,c,d}
3	{a,c,d,e}
4	{a,d,e}
5	{a,b,c}
6	{a,b,c,d}
7	{a}
8	{a,b,c}
9	{a,b,d}
10	{b,c,e}

28.14. Apply the apriori algorithm to the following data set.

Trans_id	Items_purchased
101	milk, bread, eggs
102	milk, juice
103	juice, butter
104	milk, bread, eggs
105	coffee, eggs
106	coffee
107	coffee, juice
108	milk, bread, cookies, eggs
109	cookies, butter
110	milk, bread

The set of items is {milk, bread, cookies, eggs, butter, coffee, juice}. Use 0.2 for the minimum support value.

28.15. Show two rules that have a confidence of 0.7 or greater for an itemset containing three items from Exercise 28.14.

~~**28.16.** For the partition algorithm, prove that any frequent itemset in the database must appear as a local frequent itemset in at least one partition.~~

28.17. Show the FP-tree that would be made for the data from Exercise 28.14.

28.18. Apply the FP-growth algorithm to the FP-tree from Exercise 28.17 and show the frequent itemsets.

1. Construct Decision tree from following dataset:

RID	Married	Salary	Acct_balance	Age	Loanworthy
1	no	≥50K	<5K	≥25	yes
2	yes	≥50K	≥5K	≥25	yes
3	yes	20K. . 50K	<5K	<25	no
4	no	<20K	≥5K	<25	no
5	no	<20K	<5K	≥25	no
6	yes	20K. . 50K	≥5K	≥25	yes

2. Construct Decision tree from following dataset:

RID	Age	City	Gender	Education	Repeat_customer
101	20 ... 30	NY	F	college	YES
102	20 ... 30	SF	M	graduate	YES
103	31 ... 40	NY	F	college	YES
104	51 ... 60	NY	F	college	NO
105	31 ... 40	LA	M	high school	NO
106	41 ... 50	NY	F	college	YES
107	41 ... 50	NY	F	graduate	YES
108	20 ... 30	LA	M	college	YES
109	20 ... 30	NY	F	high school	NO
110	20 ... 30	NY	F	college	YES

3. Consider the following set of two-dimensional records and Use the *k*-means, K-mediod and K-mode algorithm to cluster the data

RID	Dimension1	Dimension2
1	8	4
2	5	4
3	2	4
4	2	6
5	2	8
6	8	6

4. Consider the training examples shown in Table below for a classification problem.
- Compute the entropy of overall collection of training examples.
 - Compute the gain ratio for the Gender attribute.
 - Compute the gain ratio for the Car Type attribute
 - Compute the gain ratio for the Shirt Size attribute.

(e) Which attribute would be chosen as a root node?

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

5. Generate the cluster from following data sets using k-means algorithm (assume k=2 and no. of iterations=2):

Doc1: android, linux, android

Doc2: android, android

Doc3: java, python, python

Doc4: pascal, python, java

6. Perform agglomerative hierarchical clustering of following points:

A(0, 0), B(1,2), C(2,1), D(4,1), E(5, 0), F(5, 3)